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CPCS-214 Computer Architecture and Organization

42 - Memory Hierarchy and Technology

Introduction

- From the earliest days of computing, programmers have wanted unlimited amounts of fast memory
- Consider simple analogy that illustrates the key principles and mechanisms that we use
- Memory Hierarchy
- Memory Technology

Analogy

- Student writing term paper on computer hardware
- Sitting at a desk in library with collection of books have been pulled from shelves and are examining
- Go back to shelves and look for a book
- Having several books on desk saves time compared to having only one book there and constantly having to go back to the shelves to return it and take out another

Memory

- Same principle allows us to create illusion of large memory that we can access as fast as a very small memory
 - As no need to access all books at once with equal probability, a program does not access all of its code or data at once with equal probability
 - Otherwise, it would be impossible to make most memory accesses fast and still have large memory in computers
 - Just as it would be impossible to fit all library books on desk and still find what is wanted quickly

Principle of Locality

- Underlies both the way in which programs operate
- It states that *programs access a relatively small portion of their address space at any instant of time*
 - Just as accessing very small portion of library's collection
- Two different types of locality

Temporal Locality

- Locality in time
- If an item is referenced, it will tend to be referenced again soon
 - If you recently brought a book to your desk to look at, you will probably need to look at it again soon
- For example
 - Loops: instructions and data are likely to be accessed repeatedly

Spatial Locality

- Locality in space
- If an item is referenced, items whose addresses are close by will tend to be referenced soon
 - Libraries put books on same topic together on the same shelves to increase spatial locality
- For example
 - Instructions are normally accessed sequentially
 - Sequential accesses to elements of an array or a record will naturally have high degrees of spatial locality

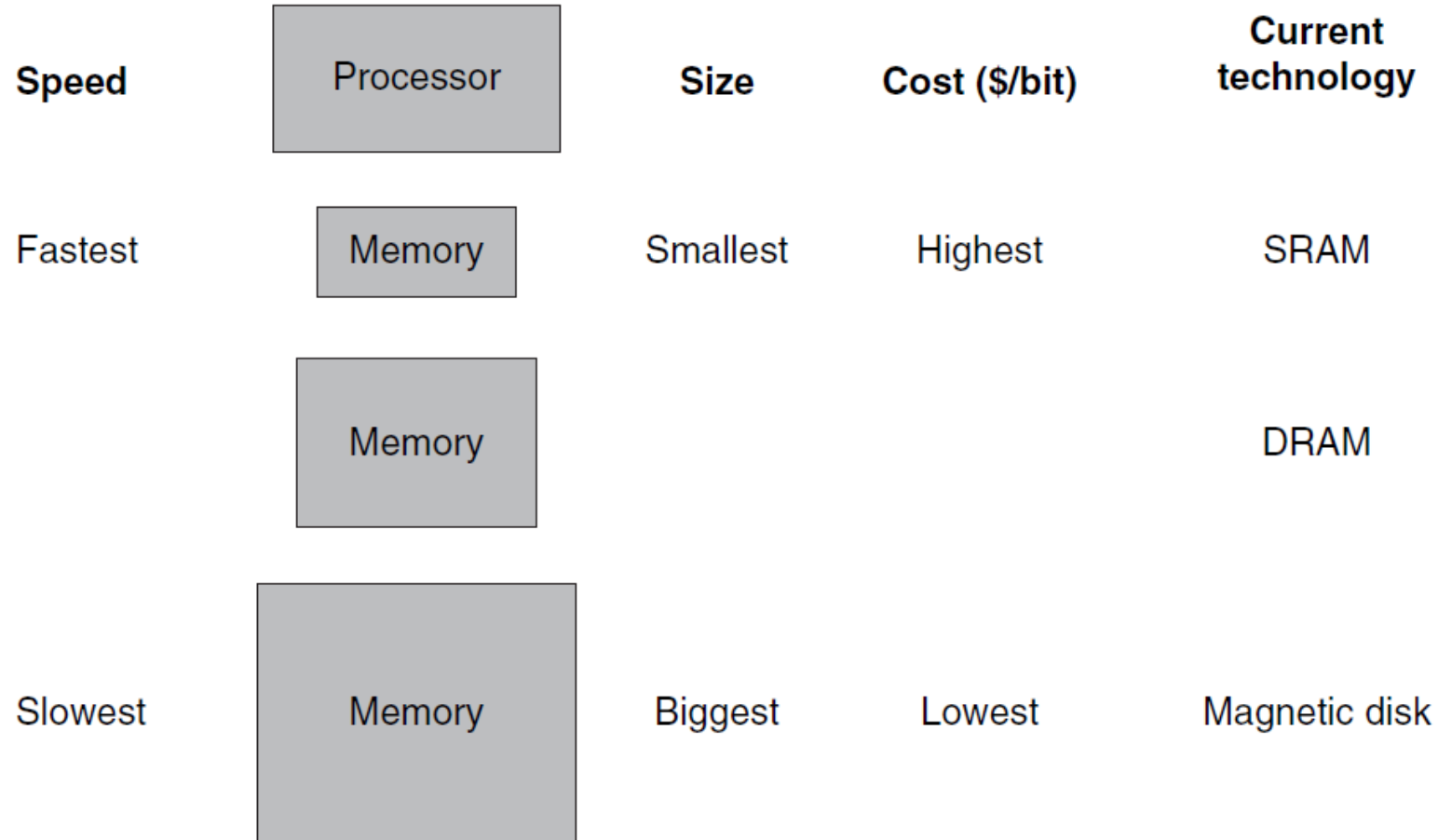
Memory Hierarchy

- Take advantage of the principle of locality by implementing memory of a computer as a **memory hierarchy**
- Consists of multiple levels of memory with different **speeds** and **sizes**
- Faster memories are more expensive per bit than slower memories and thus are smaller

Memory Hierarchy

- Figure shows faster memory is close to processor and slower, less expensive memory is below it
- Goal is to present user with as much memory as is available in the cheapest technology, while providing access at the speed offered by the fastest memory

Memory Hierarchy



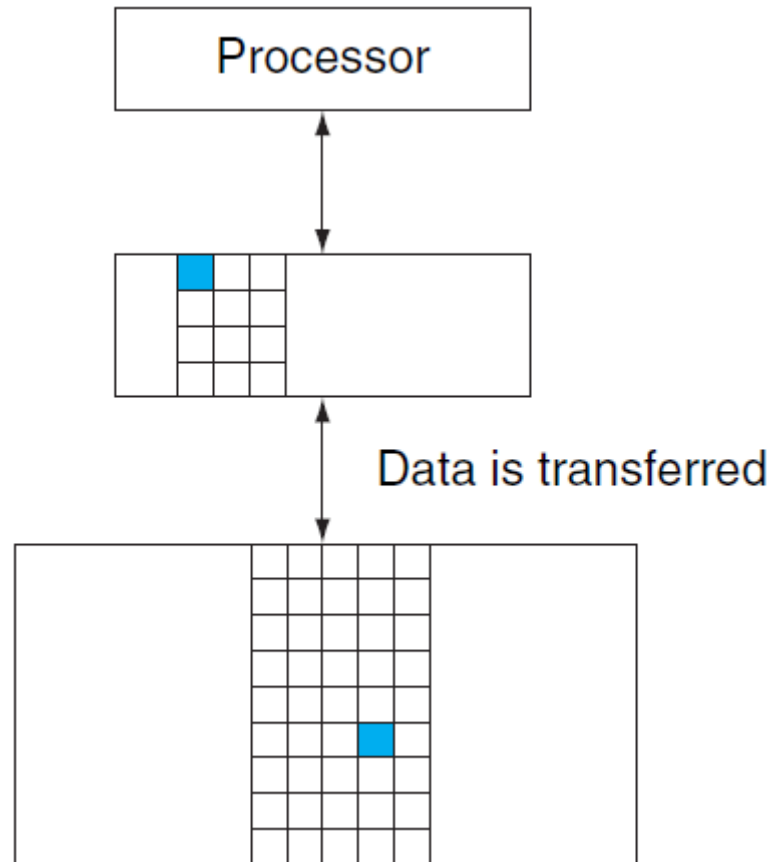
Data Hierarchy

- Data is similarly hierarchical
 - a level closer to processor is generally a subset of any level further away
 - and all data is stored at the lowest level
- Furthermore, as we move away from processor, levels take progressively longer to access
- Memory hierarchy can consist of multiple levels, but data is copied between only two adjacent levels at a time

Block

- Upper level, one closer to processor, is smaller and faster than the lower level, since upper level uses technology that is more expensive
- Minimum unit of information that can be either present or not present in the two-level hierarchy is called a **block** or a **line**
 - In library analogy, a block of information is one book

Block of Data



Hit or Miss

- If data requested by processor appears in some block in the upper level, this is called a *hit*
- If the data is not found in the upper level, the request is called a *miss*
- Lower level in hierarchy is then accessed to retrieve the block containing requested data

Hit/Miss Rate/Ratio

- **Hit rate**: fraction of memory accesses found in upper level
- Often used as a measure of performance of memory hierarchy
- **Miss rate**: fraction of memory accesses not found in upper level

$$\text{miss rate} = (1 - \text{hit rate})$$

Hit Time

- Since performance is the major reason for having a memory hierarchy, time to service hits and misses is important
- **Hit time:** time to access upper level of memory hierarchy,
 - Includes the needed to determine whether access is a hit or a miss

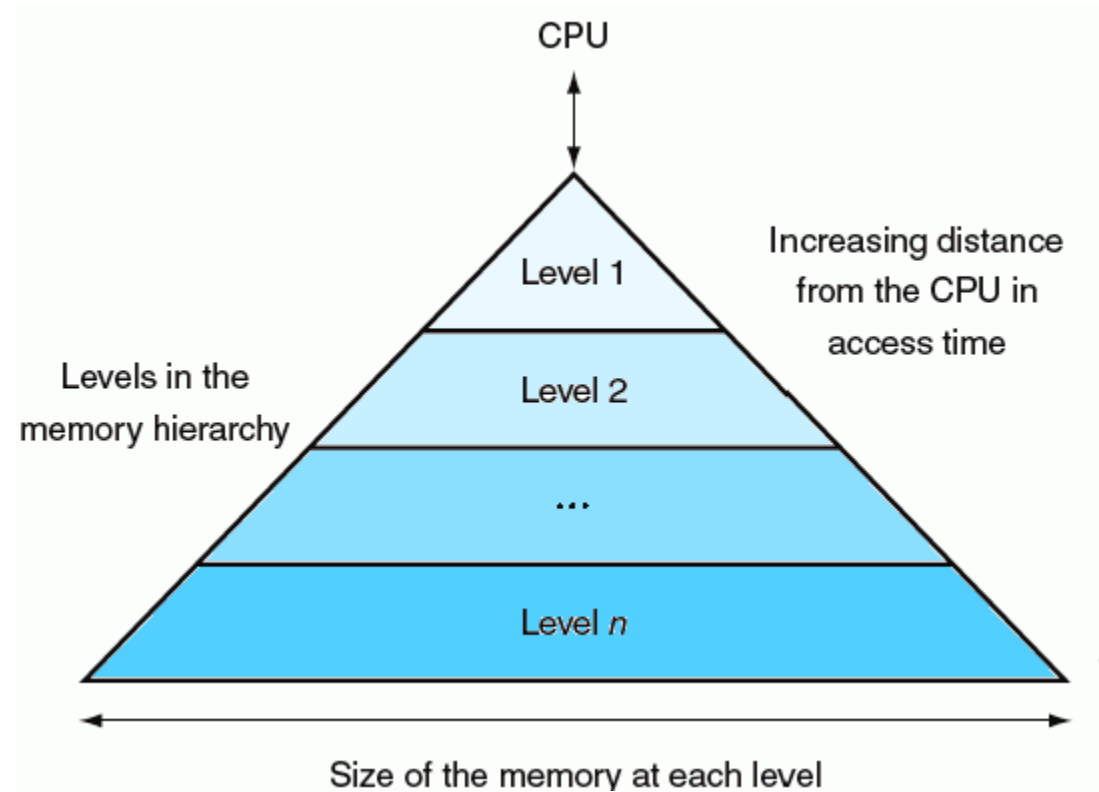
Miss Penalty

- Miss Penalty:
 - Time to replace a block in upper level with corresponding block from lower level
 - plus time to deliver this block to processor
- Because upper level is smaller and built using faster memory parts, hit time will be much smaller than time to access next level in hierarchy, which is major component of miss penalty

Memory System

- Concepts used to build memory systems affect many other aspects of a computer, including how OS manages memory and I/O, how compilers generate code, and even how applications use computer
- Because all programs spend much of their time accessing memory, the memory system is necessarily a major factor in determining performance

Structure of Memory Hierarchy



Memory Technologies

- Four primary technologies used today in memory hierarchies
- Cache
- Main memory

Cache

- Closer to the processor
- Use SRAM (static random access memory)

Main Memory

- Implemented from DRAM (dynamic random access memory),
- DRAM is less costly per bit than SRAM, although it is substantially slower
- Price difference arises because DRAM uses significantly less area per bit of memory, and DRAMs thus have larger capacity for same amount of silicon

Flash Memory

- Nonvolatile memory
- Secondary memory in PMDs

Magnetic Disk

- Used to implement the largest and slowest level in hierarchy in servers

Access Time and Price

- Access time and price per bit vary widely among these technologies, as the table below shows, using typical values for 2012

Access Time and Price

Memory technology	Typical access time	\$ per GiB in 2012
SRAM semiconductor memory	0.5–2.5 ns	\$500–\$1000
DRAM semiconductor memory	50–70 ns	\$10–\$20
Flash semiconductor memory	5,000–50,000 ns	\$0.75–\$1.00
Magnetic disk	5,000,000–20,000,000 ns	\$0.05–\$0.10